STEWARD V.A

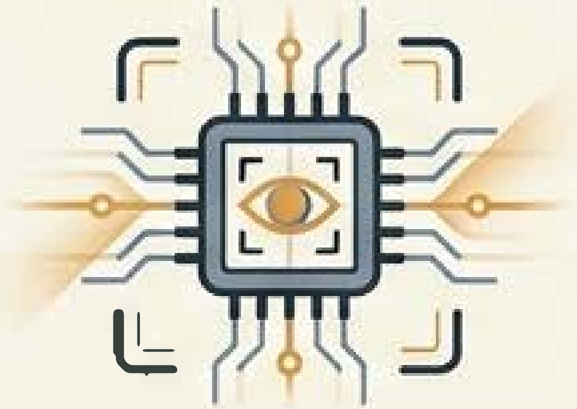
AI-Powered Independence for the Visually Impaired

Investment Opportunity | Beacon Technologies

THE OPPORTUNITY AT A GLANCE

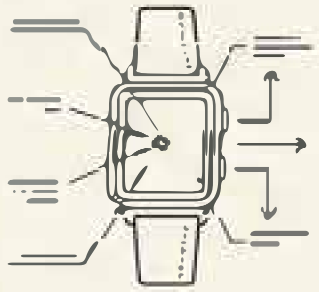
THE MISSION: BEACON TECHNOLOGIES

A Ghanaian technology company creating transformative, accessible innovations for underserved underserved communities communities via AI, IoT, and Robotics.



THE INNOVATION: STEWARD V.A

An edge-computing wearable device utilising computer vision and speech processing to deliver intelligent navigation and environmental awareness.

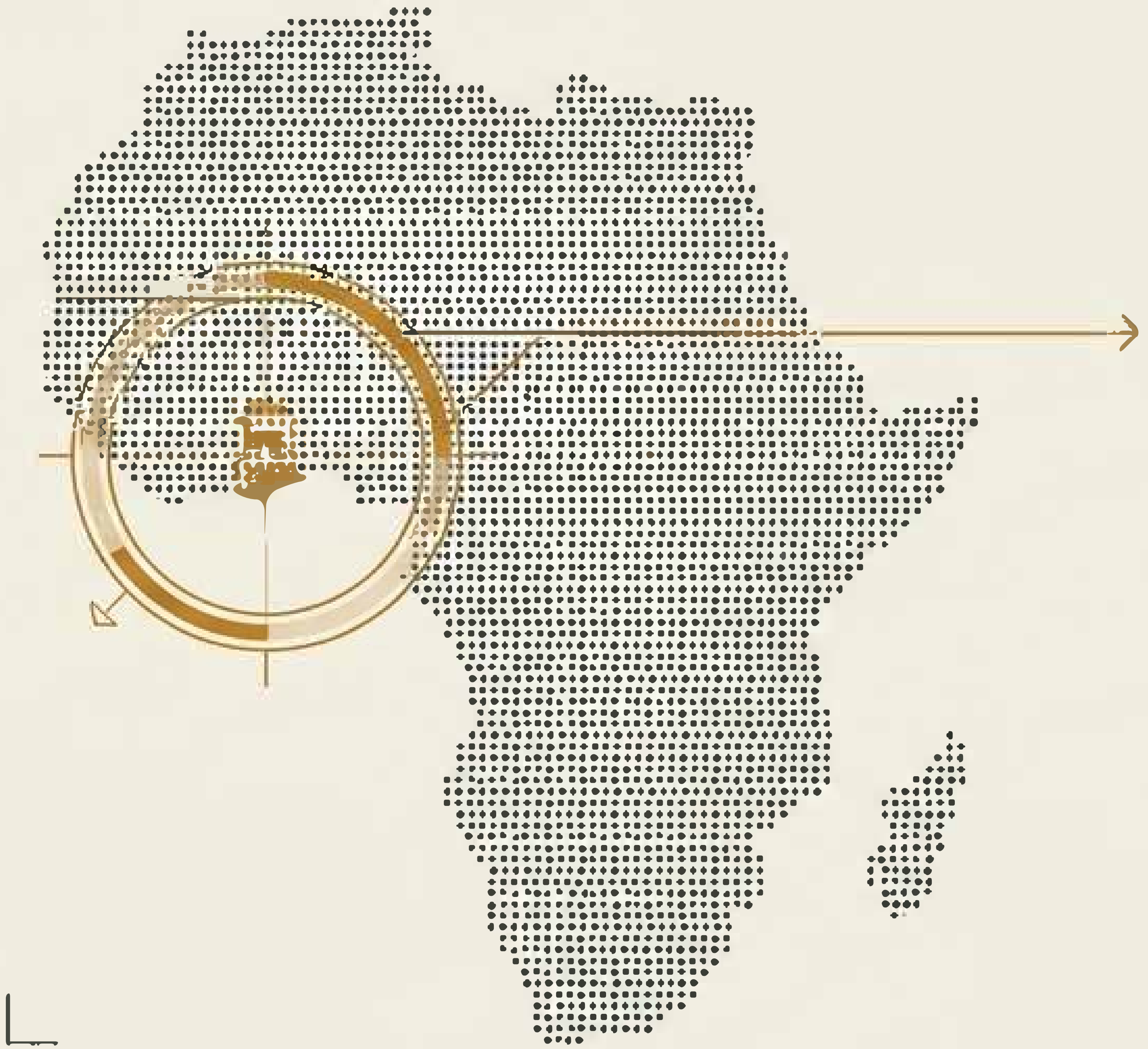


THE CATALYST: THE ASK

Seeking **US$50,000 (GHS 588,000)**

for **10% equity** to execute a 300-unit commercial pilot and cross the chasm from validated prototype to market readiness.

THE ACCESSIBILITY GAP



6 MILLION

visually impaired individuals in Africa

250,000 in Ghana alone face daily barriers to independence, education, and employment.

SYSTEM FAILURE 1: PROHIBITIVE COST

Premium assistive tech is priced for Western markets, rendering it inaccessible to the African demographic.

SYSTEM FAILURE 2: SINGLE-FUNCTION ANALOGUE

Traditional tools (like standard white canes) lack multi-functionality and environmental awareness.

SYSTEM FAILURE 3: LACKING LOCALISATION

Existing global AI solutions fail to recognise local currencies, dialects, or unique infrastructural challenges.

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MEET STEWARD V.A

OBJECT & OBSTACLE DETECTION

Real-time spatial mapping for safe navigation.

CURRENCY & TEXT RECOGNITION

Edge-computing models trained specifically for local contexts and printed materials.

VOICE INTERACTION

Real-time AI assistance and environmental auditory feedback.

AFFORDABLE. LOCALIZED. MULTI-FUNCTIONAL. DESIGNED SPECIFICALLY FOR AFRICAN ENVIRONMENTS.

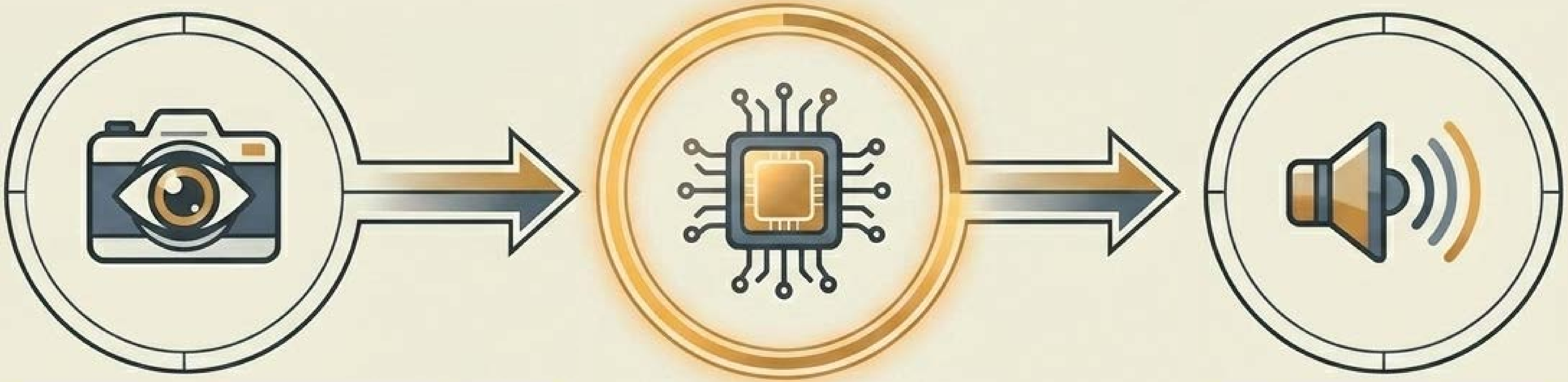
NotebookLM

THE AI PERCEPTION LOOP

1. ENVIRONMENTAL CAPTURE (INPUT)

2. EDGE COMPUTING (PROCESSING)

3. AUDITORY TRANSLATION (OUTPUT)



Sensors actively scan for physical obstacles, local currency, printed text, and spatial layout.

Proprietary computer vision and AI models process the data locally on the device, ensuring zero-latency response without requiring constant internet access.

Visual data is instantly translated into clear, contextual voice guidance and navigational cues for the user.

Turning visual chaos into structured, real-time auditory independence.

COMPETITIVE POSITIONING

Criteria	Traditional White Cane	Global AI Wearables	STEWARD V.A
Affordability for Emerging Markets	○	○	●
Multi-Functional Capabilities	○	●	●
Localised AI (e.g., Currency)	○	◔	●
Independent Edge Processing	○	◑	●

STEWARD V.A is the only solution that combines advanced, multi-functional computer vision with the economic and contextual realities of the African market.

PROVEN EFFICACY & INSTITUTIONAL BACKING

GHANA BLIND UNION

GHANA DIGITAL CENTRES LIMITED

Successful initial user validation assessing usability, accessibility, and reliability with end-users.

Awards & Recognition

- 1st Place Winner: Disability Inclusive Hackathon (DI-Hack 2024).
- Finalist: Hult Prize Ghana Nationals.
- Multiple innovation and entrepreneurship recognitions.

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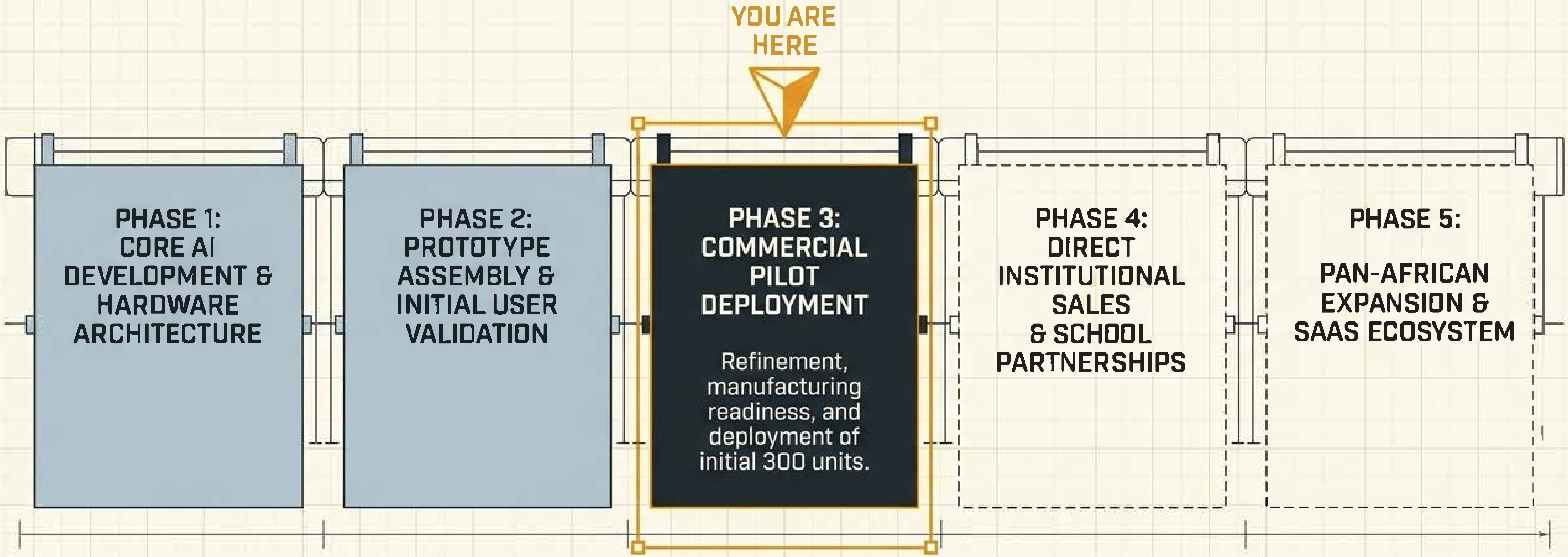
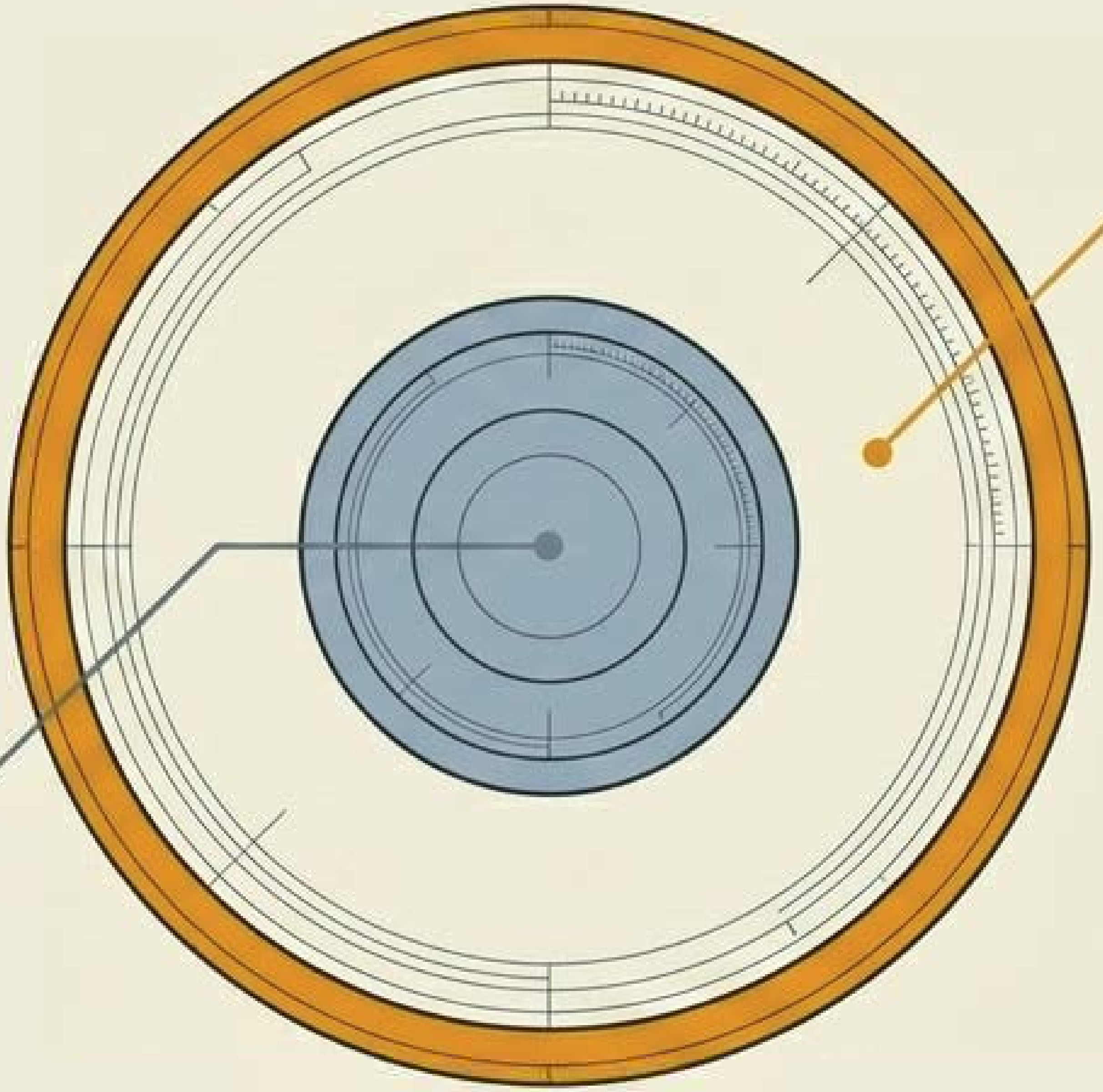
THE ROADMAP TO COMMERCIALISATION



ADDRESSABLE MARKET SCALE



Ghana Launchpad

250,000 Visually Impaired Individuals

Market Value:
GHS 625 Million

African Expansion

6 Million Visually Impaired Individuals

Market Value:
GHS 15 Billion

A deeply underserved demographic representing a highly scalable economic opportunity alongside profound social impact.

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A DIVERSIFIED REVENUE ECOSYSTEM

STEWARD V.A

INSTITUTIONAL PROCUREMENT

Bulk purchases by governments, NGOs, schools, and donor-funded programs.

DIRECT SALES & FLEXIBLE PAYMENTS

Direct-to-user sales supported by manageable instalment plans to ensure affordability.

GRANT-SUPPORTED DEPLOYMENTS

Funded accessibility and inclusion initiatives.

FUTURE SOFTWARE & SERVICES

Premium AI capabilities, advanced language modules, and ecosystem licensing.

UNIT ECONOMICS & 5-YEAR GROWTH

UNIT ECONOMICS

Production Cost: **GHS 1,300**

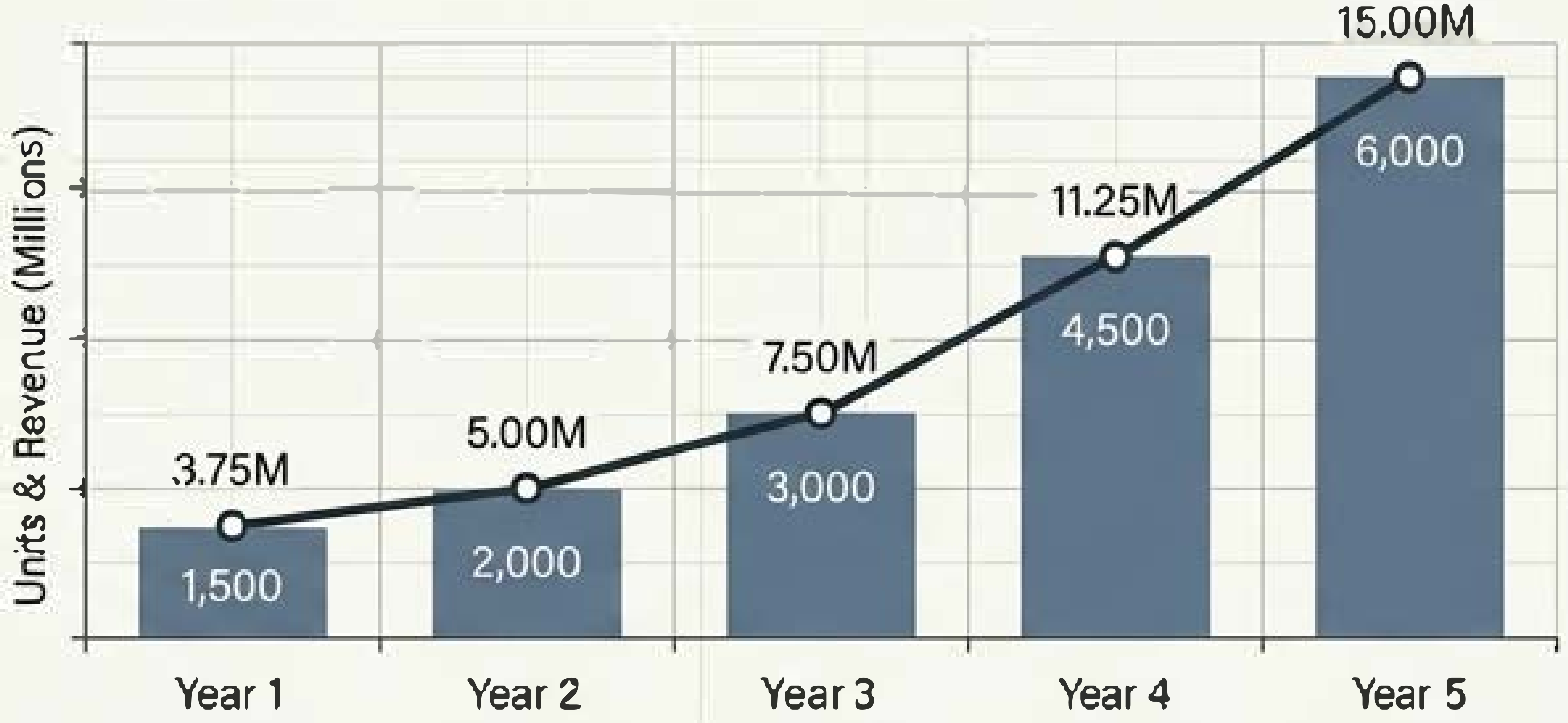
Selling Price: **GHS 2,500**

Gross Margin:

48%

GHS 1,200 Profit per unit

5-YEAR PROJECTIONS



Year 1: 1,500 units | Rev: 3.75M | Prof: 1.8M
Year 2: 2,000 units | Rev: 5.00M | Prof: 2.4M
Year 3: 3,000 units | Rev: 7.50M | Prof: 3.6M
Year 4: 4,500 units | Rev: 11.25M | Prof: 5.4M
Year 5: 6,000 units | Rev: 15.00M | Prof: 7.2M

GHS 42.5M Total 5-Year Revenue | GHS 20.4M Total Profit

THE EXECUTION ENGINE

Emmanuel Quartey Boye
Founder & Lead Innovator
Creator of STEWARD V.A, system integration, product vision.

Paul Bortey Bletcher
Technical Lead - AI/ML
Computer vision systems, AI model optimization, object recognition.

Fiagbedzi Samuel
Technical Lead - Hardware
Hardware architecture, embedded systems, sensor integration.

David Uriel Marquaye
Advisor
Strategic guidance, business development.

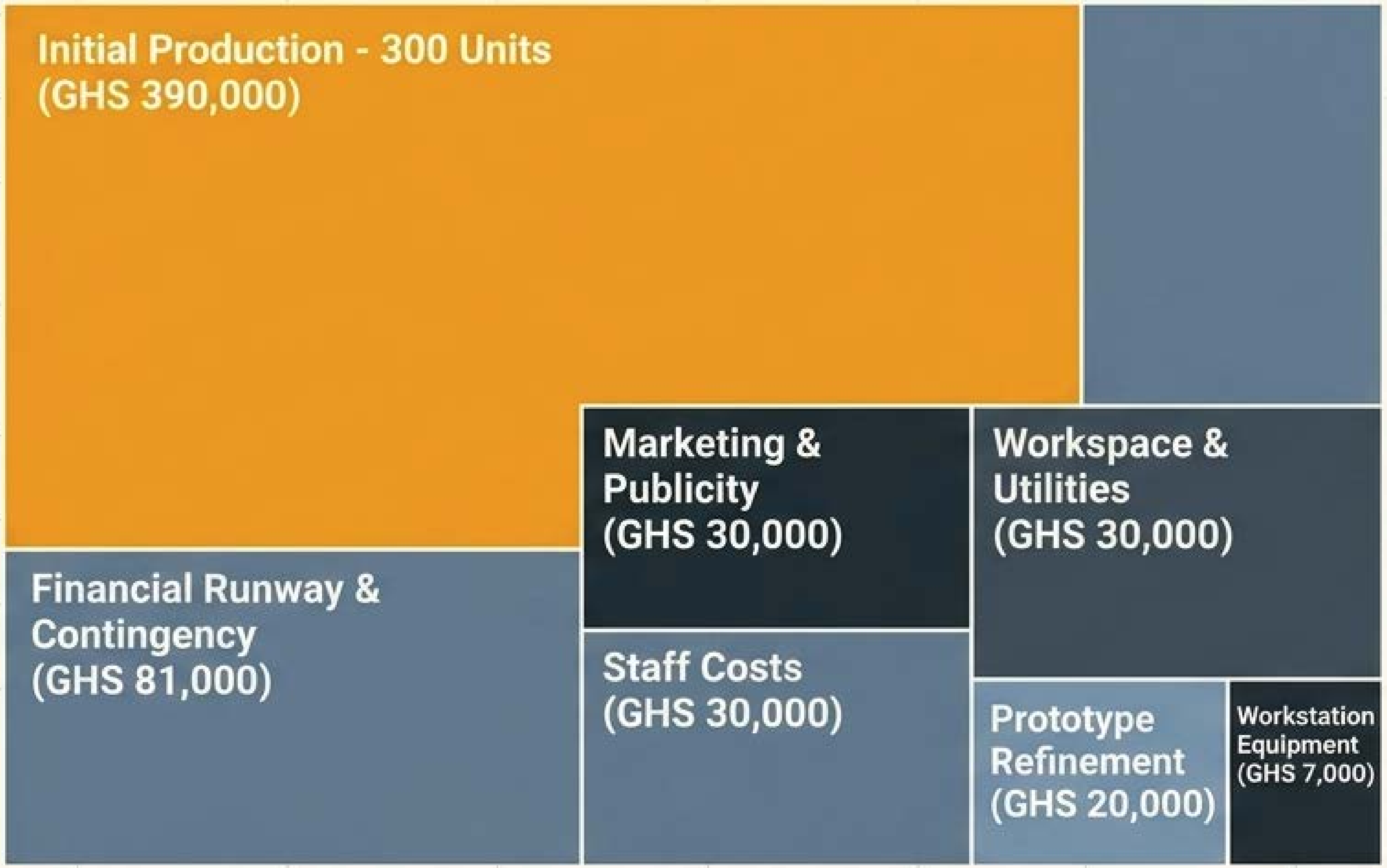
Bright Arhin Nsarko
Advisor
Industry partnerships, strategic guidance.

THE ASK: CATALYSING THE COMMERCIAL PILOT

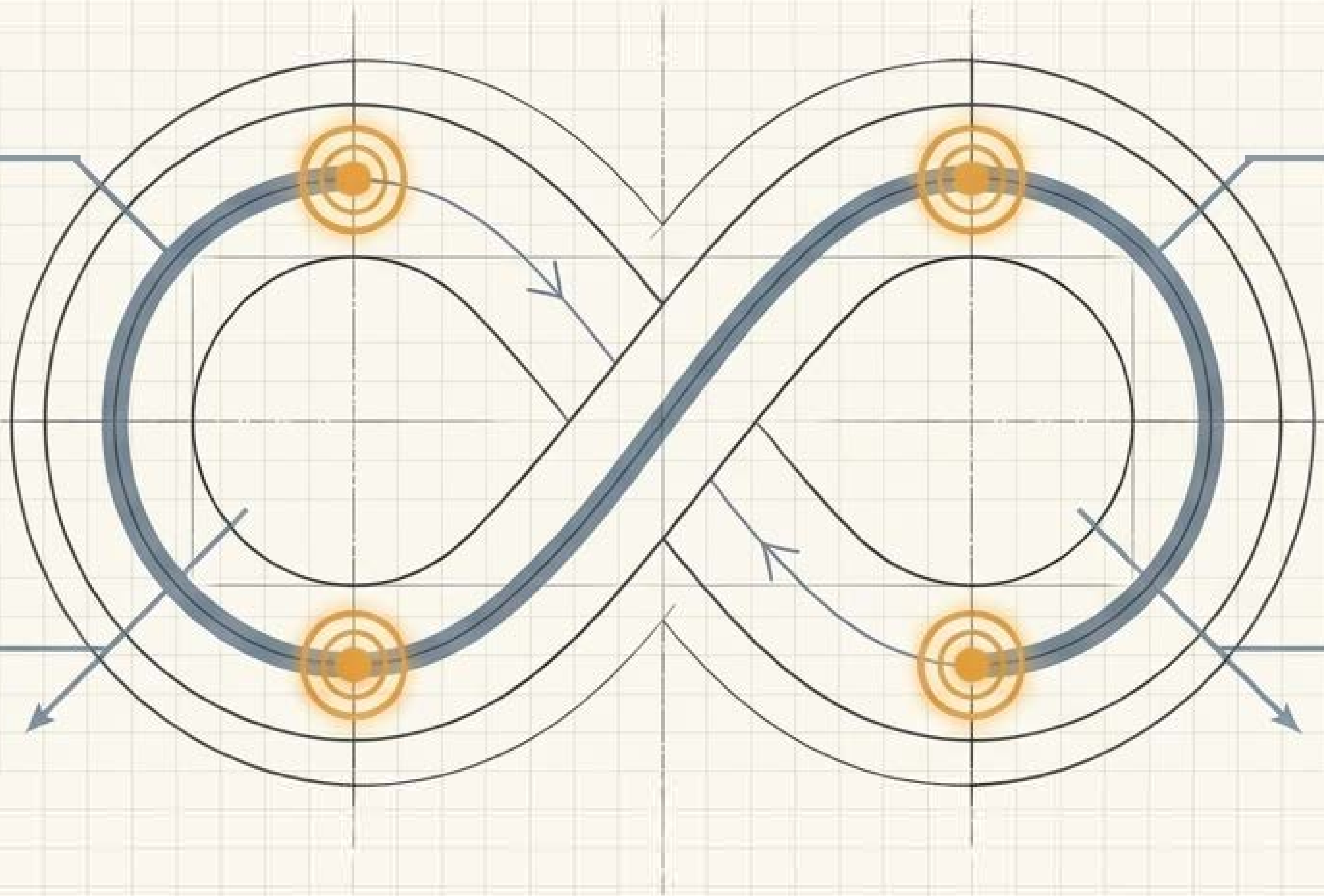
THE OFFER

US$50,000
(GHS 588,000)
for 10% Equity

USE OF FUNDS



VALUE CREATION & INVESTOR UPSIDE



PHASE 1: THE PILOT CATALYST

$50k unlocks the 300-unit pilot, securing institutional deployment partnerships and early cash flow.

PHASE 2: DATA & IP MOAT

Real-world usage generates operational data, optimising proprietary AI models and cementing intellectual property.

PHASE 3: RECURRING REVENUE SCALE

Proven outcomes unlock larger grant deployments, bulk government procurement, and consumer financing models.

PHASE 4: ENTERPRISE EXPANSION

Pan-African footprint and future product line expansion drive long-term enterprise value, opening avenues for future funding rounds.